Revealing Key Details to See Differences: A Novel Prototypical Perspective for Skeleton-based Action Recognition

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Abstract

Skeleton-based action recognition faces a fundamental challenge: distinguishing visually similar actions that differ only in subtle joint-level motion patterns. Existing graph convolutional networks (GCNs) tend to capture coarse body dynamics and often miss fine-grained discriminative details concentrated in specific joints. In this paper, we propose ***ProtoGCN****, a novel framework that models skeleton dynamics as combinations of reusable motion prototypes. At its core, a Prototype Reconstruction Network (PRN) maintains a learnable prototype memory and uses attention-based addressing to reconstruct graph topology features into enriched, discriminative representations. A Motion Topology Enhancement (MTE) module injects nuanced intra-sample and inter-sample topology signals into each GCN layer, while a Class-Specific Contrastive Learning (CSCL) objective drives inter-class separation via a momentum-updated class center memory bank. Extensive experiments show that ProtoGCN achieves state-of-the-art performance on NTU RGB+D (93.8% X-Sub, 97.8% X-View), NTU RGB+D 120 (90.9% X-Sub, 92.2% X-Set), Kinetics-Skeleton (51.9% Top-1), and FineGYM (95.9%), validating the effectiveness of prototypical motion representations for fine-grained skeleton-based action recognition.*

1. Introduction

Human action recognition from skeleton sequences is a central problem in computer vision, with applications in surveillance, healthcare, sports analysis, and human-computer interaction. Skeleton data offers compelling advantages over RGB video: it is compact, invariant to appearance changes, and robust to viewpoint variation. These properties have spurred a rich body of work on graph convolutional networks (GCNs) that model the human body as a spatial graph with joints as nodes and bones as edges [2, 14, 16, 24].

Despite remarkable progress, a critical challenge remains: distinguishing actions that share similar overall body configurations but differ only in fine-grained joint dynamics. As illustrated in Figure 1, actions such as *Writing* and *Typing on a Keyboard* involve nearly identical global postures, yet their discriminative cues are confined to subtle wrist and finger joint relationships. Existing GCNs learn globally distributed graph topologies that spread representational capacity evenly across joints, diluting the fine-grained signals that matter most for such confusable actions. This limitation is compounded by intra-class variation—different subjects performing the same action differently—and noise in pose estimation, which further obscures the subtle patterns.

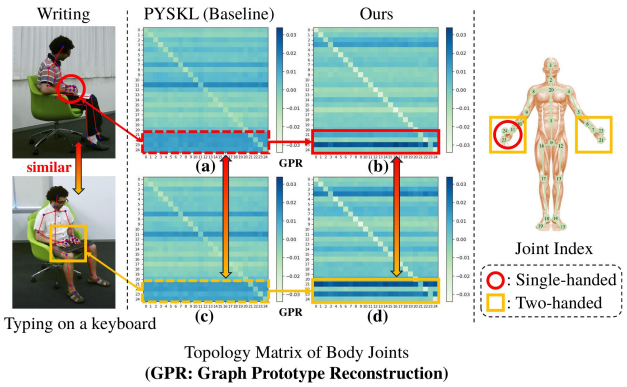

Figure 1. Illustration of the skeletons and learned topologies for similar actions *Writing* and *Typing on a Keyboard* (deeper color indicates stronger relationships between corresponding joints). ProtoGCN learns focused, discriminative joint relationships that reveal key differences between confusable actions.

We propose a new paradigm: model action dynamics as *combinations of reusable motion prototypes*—compact templates encoding characteristic joint-relationship patterns shared across the training set. Given an input skeleton sequence, its topology is reconstructed by retrieving and composing relevant prototypes, inherently emphasizing the joint patterns most diagnostic for the particular motion type. This prototype-based perspective naturally focuses attention on discriminative details and promotes generalization by sharing learned motion primitives across samples and classes.

Based on this insight, we present **ProtoGCN**, whose architecture is shown in Figure 2. ProtoGCN integrates three tightly coupled innovations:

- **Prototype Reconstruction Network (PRN):** A learnable prototype memory bank with softmax attention-based addressing that maps input topology features to reconstructed representations emphasizing discriminative joint-relationship patterns.
- **Motion Topology Enhancement (MTE):** A module embedded in each GCN layer that computes intra-sample joint correlations ($\mathbf{A}_{\text{intra}}$) and inter-sample joint distinctions ($\mathbf{A}_{\text{inter}}$) to enrich the adjacency matrix used for feature aggregation and prototype retrieval.
- **Class-Specific Contrastive Learning (CSCL):** A contrastive objective with a momentum-updated class center memory bank that enforces intra-class compactness and inter-class separation in the feature space.

ProtoGCN achieves state-of-the-art results on four standard benchmarks, with the most pronounced gains concentrated on the hardest-to-distinguish action categories (up to +3.0% on the 10 most difficult classes in NTU-120).

Our contributions are as follows:

- We propose ProtoGCN, a prototype-based GCN that reconstructs motion topology features from a learnable prototype memory, revealing fine-grained discriminative joint dynamics.
- We design the PRN with differentiable softmax addressing over a learnable prototype matrix, enabling input-adaptive topology reconstruction end-to-end.
- We introduce MTE to enrich each GCN layer with $\mathbf{A}_{\text{intra}}/\mathbf{A}_{\text{inter}}$ topology signals, and CSCL to enforce class-level feature compactness and dispersion via momentum-updated class centers.
- ProtoGCN sets new state-of-the-art on NTU RGB+D, NTU RGB+D 120, Kinetics-Skeleton, and FineGYM, with thorough ablations confirming each component's contribution.

2. Related Work

Skeleton-based Action Recognition with GCNs. The pioneering ST-GCN [24] demonstrated that the human body can be naturally modeled as a spatial-temporal graph, enabling effective action recognition from skeleton sequences. Subsequent work has expanded this foundation in multiple directions. AS-GCN [12] introduced actional and structural link mining to capture non-physical joint dependencies beyond the anatomical skeleton. 2s-AGCN [16] proposed adaptive graph topology learning using attention mechanisms. MS-G3D [14] disentangled multi-scale temporal and spatial dependencies. DC-GCN+ADG [4] decoupled graph convolutions for richer feature extraction. MST-GCN [3] integrated multi-scale spatial-temporal modeling. CTR-GCN [2] introduced channel-wise topology refinement, allowing different channels to attend to different joint relationships. InfoGCN [5] leveraged information-theoretic objectives for richer representations. More recent advances include HD-GCN [11] with hierarchical body-part decomposition, DS-GCN [20] with dynamic semantic graph construction, and BlockGCN [27] with block-wise graph decomposition. Methods such as PYSKL [7], STF [10], FR-Head [26], GAP [19], and JT-GraphFormer [25] further push performance on standard benchmarks.

Despite these advances, most methods learn globally distributed topology representations that do not explicitly focus on the small joint subsets carrying the most discriminative information. ProtoGCN directly addresses this by decomposing topology features into prototype-driven representations that highlight the joint relationships most relevant to each input.

Contrastive and Prototype Learning. Contrastive learning has emerged as a powerful paradigm for discriminative representation learning [9, 26]. SkeletonGCL [9] applies graph contrastive learning to skeleton data by encouraging consistency across augmented views; FR-Head [26] introduces frequency-domain contrastive heads. Our CSCL extends the prototype learning perspective with a momentum-updated class center bank, providing stable class-specific anchors for contrastive supervision. Unlike instance-level contrastive methods, CSCL operates at the class level, directly enforcing inter-class discriminability.

Transformer-based Skeleton Recognition. Transformers have emerged as strong alternatives to GCNs for skeleton-based action recognition. SkateFormer [6] proposes a skeletal-temporal transformer that partitions joints and frames by relation type and applies partition-specific self-attention, achieving state-of-the-art results while reducing memory overhead. While these methods demonstrate the power of global self-attention for capturing non-local joint dependencies, they typically require larger model capacity. ProtoGCN retains the structural inductive bias of graph convolution while augmenting it with prototype-driven topology reconstruction, achieving competitive performance with a more interpretable and parameter-efficient design.

Memory-Augmented Networks and Prototype Learning. Memory-augmented neural networks [15] have demonstrated the ability to store and retrieve persistent knowledge representations, enabling models to go beyond what fits in standard network parameters. Modern associative memory models provide exponential storage capacity for prototype-like patterns, suggesting the viability of memory-based reconstruction for complex recognition tasks. In the context of few-shot learning, prototype-based classifiers [17] show that aggregated class embeddings serve as effective and compact proxies for recognition decisions. In action recognition, however, prototype-

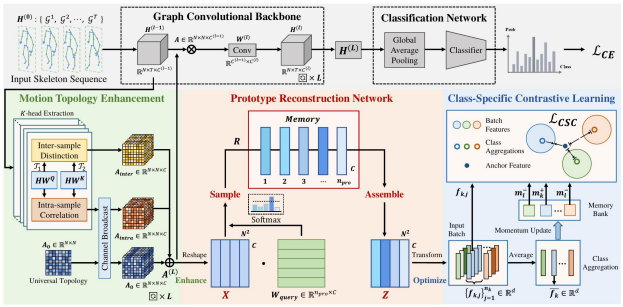

Figure 2. The overall architecture of ProtoGCN. The backbone GCN with Motion Topology Enhancement (MTE) at each layer produces topology features that are reconstructed by the Prototype Reconstruction Network (PRN). Class-Specific Contrastive Learning (CSCL) acts on the reconstructed representation during training.

based approaches [22] have remained underexplored relative to their potential. While some prior works apply external memory modules to video understanding for capturing long-range temporal dependencies, they do not address the fine-grained, topology-level pattern decomposition that is central to our approach. ProtoGCN bridges these threads by introducing a differentiable prototype memory that decomposes graph topology features into reusable motion building blocks, enabling input-adaptive discriminative reconstruction without requiring a separate clustering or pre-training stage.

Video and Pose-based Action Recognition. For the FineGYM benchmark we also compare against RGB-based approaches including I3D [1], TSN [18], TSM [13], and LT-S3D [21], as well as skeleton-based methods PoseConv3D [8] and SkeletonMAE [23], demonstrating that ProtoGCN achieves superior results using only skeleton data.

3. Method

Figure 2 provides an overview of ProtoGCN. The backbone consists of 10 GCN-TCN stages processing skeleton joint coordinate sequences. Channel widths are inflated at stages 5 and 8, and temporal downsampling is applied at stages 5 and 8. Each GCN layer is augmented with MTE. After the final GCN stage, the graph topology output is passed to the PRN for prototype-based reconstruction. CSCL is applied during training using the reconstructed graph representation. We describe each component in detail below.

3.1. Motion Topology Enhancement (MTE)

Given an input feature map $\mathbf{x} \in \mathbb{R}^{N\times C\times T\times V}$ (batch size N, channels C, frames T, joints V), each unit_gcn module computes two auxiliary projections $\mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^{N\times K\times C'\times T\times V}$ (for K graph subsets and C' mid-channels) via separate 1×1 convolutions. Temporal mean pooling reduces these to $N\times K\times C'\times 1\times V$.

Inter-sample topology $\mathbf{A}_{\text{inter}}$ captures pairwise joint distinctions within a sample:

$$\mathbf{A}_{\text{inter}} = \alpha \cdot \tanh\big(\mathbf{x}_1 - \mathbf{x}_2^{\top}\big) \in \mathbb{R}^{N\times K\times C'\times V\times V}, \quad (1)$$

where α is a learnable scalar and the $\tanh$ activation allows signed joint interaction strengths.

Intra-sample topology $\mathbf{A}_{\text{intra}}$ captures joint co-activation patterns:

$$\mathbf{A}_{\text{intra}} = \beta \cdot \mathrm{softmax}(\mathbf{x}_1 \otimes \mathbf{x}_2) \in \mathbb{R}^{N\times K\times C'\times V\times V}, \quad (2)$$

where β is a learnable scalar and $\otimes$ denotes the outer product across joints, normalized by softmax so attention weights sum to 1 over neighbors.

The enhanced adjacency used for graph convolution is:

$$\mathbf{A} = \mathbf{A}_{\text{base}} + \mathbf{A}_{\text{inter}} + \mathbf{A}_{\text{intra}}, \quad (3)$$

where $\mathbf{A}_{\text{base}}$ is the learnable base adjacency matrix. This composite adjacency enriches the graph topology used for both feature aggregation and downstream prototype retrieval.

3.2. Prototype Reconstruction Network (PRN)

After the backbone extracts graph topology features $\mathbf{G} \in \mathbb{R}^{N\times C\times V\times V}$ from the last GCN layer, the PRN reconstructs them using a learnable prototype memory. We reshape $\mathbf{G}$ as $\mathbf{X} \in \mathbb{R}^{N\times V^2\times C}$, where each of the V^2 joint-pair positions is represented by a C-dimensional feature vector.

Prototype memory addressing: A query matrix $\mathbf{W}_q \in \mathbb{R}^{C\times n_{\text{pro}}}$ (where n_{pro} is the number of prototypes) projects each feature to attention weights over the prototype bank:

$$\mathbf{Q} = \mathrm{softmax}(\mathbf{X}\,\mathbf{W}_q) \in \mathbb{R}^{N\times V^2\times n_{\text{pro}}}. \quad (4)$$

The softmax ensures each joint-pair position retrieves a convex combination of prototypes.

Prototype assembly: A memory matrix $\mathbf{W}_m \in \mathbb{R}^{n_{\text{pro}}\times C}$ assembles the retrieved prototypes into an enhanced representation:

$$\mathbf{Z} = \mathbf{Q}\,\mathbf{W}_m \in \mathbb{R}^{N\times V^2\times C}, \quad (5)$$

followed by dropout regularization. $\mathbf{Z}$ is reshaped to $N\times C\times V\times V$, refined by a 1×1 convolution, batch normalization, and ReLU, and then globally averaged over channels to obtain a compact graph representation $\mathbf{r} \in \mathbb{R}^{N\times V^2}$.

Both $\mathbf{W}_q$ and $\mathbf{W}_m$ are learned end-to-end. The prototype memory (encoded in $\mathbf{W}_m$) is shared across all samples, allowing the model to discover recurring motion building blocks. By default we use $n_{\text{pro}} = 100$ and $C = 384$.

3.3. Class-Specific Contrastive Learning (CSCL)

To drive inter-class discriminability in the reconstructed representation space, CSCL applies a supervised contrastive objective anchored to per-class feature centers. A projection head $\mathbf{W}_{cl} \in \mathbb{R}^{V^2 \times d}$ ($d = 256$) maps $\mathbf{r}$ to a compact embedding $\mathbf{f} = \mathbf{r}\,\mathbf{W}_{cl} \in \mathbb{R}^{N \times d}$.

Class center memory bank: CSCL maintains a momentum-updated bank $\mathbf{M} \in \mathbb{R}^{d \times K}$ (K classes). At each step, correctly predicted training samples update their class center:

$$\mathbf{m}_k \leftarrow m\,\mathbf{m}_k + (1 - m)\,\bar{\mathbf{f}}_k, \quad m = 0.9, \tag{6}$$

where $\bar{\mathbf{f}}_k$ is the batch-mean embedding of correctly predicted samples for class k.

Contrastive loss: With ℓ_2-normalized features and class centers, CSCL computes a cross-entropy over cosine similarity scores at temperature $\tau = 0.125$:

$$\mathcal{L}_{\text{CSCL}} = -\frac{1}{N} \sum_{i=1}^{N} \log \frac{\exp(\mathbf{f}_i \cdot \mathbf{m}_{y_i}/\tau)}{\sum_{k=1}^{K} \exp(\mathbf{f}_i \cdot \mathbf{m}_k/\tau)}. \tag{7}$$

Total loss: The final training objective combines standard cross-entropy classification with CSCL:

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda\,\mathcal{L}_{\text{CSCL}}. \tag{8}$$

This joint objective simultaneously optimizes recognition accuracy and inter-class feature dispersion. The weight λ is a hyperparameter analyzed in Figure 3.

3.4. Training and Inference

During training, ProtoGCN jointly optimizes the backbone GCN layers, the MTE modules, the PRN parameters $\mathbf{W}_q$ and $\mathbf{W}_m$, and the CSCL projection head $\mathbf{W}_{cl}$ end-to-end via backpropagation. The prototype memory $\mathbf{W}_m$ is randomly initialized and then updated entirely through gradient descent, allowing the model to discover recurring motion building blocks directly from the training signal without any pre-clustering step. The class center memory bank $\mathbf{M}$ is initialized to zero and populated via momentum updates after each training iteration; because no gradients flow through $\mathbf{M}$, it acts as a stable, slowly drifting set of class anchors that prevents collapse in the contrastive objective. During inference, the CSCL projection head and the memory bank are discarded entirely; only the backbone with MTE and the PRN are active. The final class prediction is obtained by a linear classifier applied to the global average pooled PRN representation, which is optionally aggregated across multiple input streams in the multi-stream ensemble. This lightweight inference pipeline keeps ProtoGCN practical for deployment while retaining all recognition gains from prototype-based reconstruction.

4. Experiments

4.1. Datasets

We evaluate ProtoGCN on four widely used benchmarks for skeleton-based action recognition.

NTU RGB+D [24] contains 56,880 video samples across 60 action classes performed by 40 subjects captured from three camera viewpoints. We follow the two standard evaluation protocols: Cross-Subject (X-Sub), which splits subjects into training and testing groups, and Cross-View (X-View), which uses different camera viewpoints for training and testing.

NTU RGB+D 120 [2] extends the above dataset to 114,480 samples covering 120 action classes with 106 subjects and 32 camera setups. Evaluation follows Cross-Subject (X-Sub) and Cross-Setup (X-Set) protocols.

Kinetics-Skeleton [24] is derived from the large-scale Kinetics-400 video dataset by estimating 2D skeleton coordinates using OpenPose. It contains approximately 240,000 training clips and 20,000 validation clips spanning 400 human action categories.

FineGYM [7] is a fine-grained action recognition dataset consisting of gymnastic routines that require distinguishing between subtle and visually similar movements, making it particularly suitable for evaluating methods designed to capture discriminative motion details.

4.2. Implementation Details

We adopt CTR-GCN [2] as the backbone for our ProtoGCN framework. Following prior work [5, 11], we employ a multi-stream ensemble strategy using six input modalities: joint positions, bone vectors, joint motion, bone motion, joint acceleration, and bone acceleration. Models are trained with SGD using an initial learning rate of 0.1, weight decay of 4×10^{-4}, and a cosine annealing schedule over 110 epochs, with a linear warm-up period of 10 epochs and a batch size of 64. We apply standard skeleton data augmentations including random rotation, random shear, and random scaling during training. Dropout with rate 0.25 is applied after the PRN. For the MTE module, the number of graph subsets is set to $K = 8$ and the mid-channel dimension to $C' = 64$. All experiments are conducted on 4 NVIDIA A100 GPUs. The contrastive loss weight λ and prototype memory capacity n_{pro} are analyzed in Figs. 3 and 4. All accuracy comparisons are conducted on the NTU RGB+D 120 Cross-Subject benchmark unless stated otherwise.

4.3. Comparison with State-of-the-Art

Tab. 1 presents comprehensive comparisons against state-of-the-art methods on NTU RGB+D, NTU RGB+D 120, and Kinetics-Skeleton. With the 6-stream ensemble, ProtoGCN achieves 93.8% and 97.8% on NTU

Table 1. Performance comparisons against state-of-the-art methods on the NTU RGB+D, NTU RGB+D 120, and Kinetics-Skeleton datasets in terms of classification accuracy (%).

Methods	Publication	NTU RGB+D		NTU RGB+D 120		Kinetics-Skeleton	
		X-Sub	X-View	X-Sub	X-Set	Top-1	Top-5
ST-GCN [24]	AAAI 2018	81.5	88.3	70.7	73.2	30.7	52.8
AS-GCN [12]	CVPR 2019	86.8	94.2	78.3	79.8	34.8	56.5
2s-AGCN [16]	CVPR 2019	88.5	95.1	82.5	84.2	36.1	58.7
MS-G3D [14]	CVPR 2020	91.5	96.2	86.9	88.4	38.0	60.9
DC-GCN+ADG [4]	ECCV 2020	90.8	96.6	86.5	88.1	-	-
MST-GCN [3]	AAAI 2021	91.5	96.6	87.5	88.8	38.1	60.8
CTR-GCN [2]	ICCV 2021	92.4	96.8	88.9	90.6	-	-
STF [10]	AAAI 2022	92.5	96.9	88.9	89.9	39.9	-
InfoGCN [5]	CVPR 2022	93.0	97.1	89.8	91.2	-	-
PYSKL [7]	ACM MM 2022	92.6	97.4	88.6	90.8	49.1	-
SkeletonGCL [9]	ICLR 2023	92.8	97.1	89.8	91.2	-	-
FR-Head [26]	CVPR 2023	92.8	96.8	89.5	90.9	-	-
GAP [19]	ICCV 2023	92.9	97.0	89.9	91.1	-	-
HD-GCN [11]	ICCV 2023	93.4	97.2	90.1	91.6	40.9	63.5
JT-GraphFormer [25]	AAAI 2024	93.4	97.5	89.9	91.7	-	-
DS-GCN [20]	AAAI 2024	93.1	97.5	89.2	91.1	50.6	-
BlockGCN [27]	CVPR 2024	93.1	97.0	90.3	91.5	-	-
ProtoGCN (2-ensemble)		93.0	97.2	89.7	91.2	49.9	74.0
ProtoGCN (4-ensemble)		93.5	97.5	90.4	91.9	51.3	75.1
ProtoGCN (6-ensemble)		**93.8**	**97.8**	**90.9**	**92.2**	**51.9**	**75.6**

Table 2. Performance comparisons against state-of-the-art methods on FineGYM in terms of classification accuracy (%).

Methods	Modality	FineGYM
I3D [1]	RGB	64.4
TSN [18]	RGB+Flow	79.8
TSM [13]	RGB+Flow	81.2
LT-S3D [21]	RGB	88.9
SkeletonMAE [23]	Skeleton	91.8
PYSKL [7]	Skeleton	94.1
PoseConv3D [8]	Skeleton+Limb	94.3
ProtoGCN (Ours)	Skeleton	**95.9**

RGB+D under X-Sub and X-View settings, surpassing JT-GraphFormer [25] by +0.4% and +0.3%, respectively. On the more challenging NTU RGB+D 120, ProtoGCN obtains 90.9% (X-Sub) and 92.2% (X-Set), outperforming Block-GCN [27] by +0.6% and +0.7%. On Kinetics-Skeleton, our method achieves 51.9% Top-1 and 75.6% Top-5 accuracy, improving upon DS-GCN [20] by +1.3% in Top-1 accuracy. Notably, even the 2-stream ensemble already achieves competitive performance with many existing methods, demonstrating that the gains stem from our proposed modules rather than merely from the ensemble strategy.

Table 3. Ablation study on the contribution of each component in ProtoGCN under the NTU-120 X-Sub setting.

Baseline	MTE	PRN	CSCL	Acc (%)
✓	–	–	–	87.8
✓	✓	–	–	88.1 (↑ 0.3)
✓	✓	–	✓	88.3 (↑ 0.5)
✓	✓	✓	–	88.5 (↑ 0.7)
✓	✓	✓	✓	**89.0 (↑ 1.2)**

We further evaluate ProtoGCN on FineGYM (Tab. 2), a fine-grained gymnastic action recognition benchmark. ProtoGCN achieves 95.9%, outperforming skeleton-based methods PYSKL [7] (94.1%) and PoseConv3D [8] (94.3%) by clear margins. This result confirms the effectiveness of prototype-based reconstruction in capturing subtle motion differences between visually similar actions.

4.4. Ablation Studies

Component analysis. Tab. 3 presents the contribution of each proposed component. Starting from the baseline (87.8%), adding the Motion Topology Enhancement (MTE) module yields +0.3%, indicating that the enriched topology signals provide useful complementary information. In-

Table 4. Comparison of classification accuracies (%) based on (*left*) existing graph contrastive methods and (*right*) different enhanced topologies for the network.

Methods	Acc (%)	Methods	Acc (%)
Baseline	87.8	ProtoGCN	**89.0**
w/ SkeletonGCL [9]	88.0	w/o $\mathbf{A}_{\mathbf{intra}}$	88.8
w/ FR-Head [26]	88.2	w/o $\mathbf{A}_{\mathbf{inter}}$	88.5
Ours	**89.0**	w/o $\mathbf{A}_{\mathbf{intra}}, \mathbf{A}_{\mathbf{inter}}$	88.4

Table 5. Average performance (%) on different difficulty level classes sorted by accuracy under the NTU-120 X-Sub setting.

Difficulty Level	Baseline	ProtoGCN	Δ
class 1-10	64.1	67.1	**+3.0**
class 11-30	78.6	81.1	**+2.5**
class 31-60	88.6	89.7	**+1.1**
class 61-120	96.0	96.4	**+0.4**

corporating Class-Specific Contrastive Learning (CSCL) on top of MTE brings a further +0.2%, while adding the Prototype Reconstruction Network (PRN) instead yields +0.4%, suggesting that prototype-based reconstruction contributes more individually than the contrastive objective alone. Combining all three components achieves the best performance of 89.0% (+1.2%), demonstrating their complementary nature.

Comparison with graph contrastive methods. The left portion of Tab. 4 compares ProtoGCN against existing graph contrastive learning approaches. SkeletonGCL [9] and FR-Head [26] improve the baseline to 88.0% and 88.2%, respectively, while our full approach achieves 89.0%, demonstrating the superiority of the prototype-based paradigm over conventional graph contrastive strategies.

Topology ablation. The right portion of Tab. 4 examines the contribution of the enhanced topology components. Removing $\mathbf{A}_{\mathbf{intra}}$ reduces accuracy to 88.8%, while removing $\mathbf{A}_{\mathbf{inter}}$ causes a larger drop to 88.5%, indicating that inter-sample distinction signals are particularly valuable. Removing both components degrades performance to 88.4%, confirming that both topology enhancements contribute meaningfully.

Difficulty-level analysis. Tab. 5 reports average accuracy across action classes grouped by difficulty level. ProtoGCN achieves the largest improvement on the hardest classes (class 1–10): +3.0% over the baseline. The gain decreases for progressively easier classes (+2.5%, +1.1%, +0.4%), confirming that prototype-based reconstruction specifically benefits challenging actions where fine-grained discrimination is most critical.

Hyperparameter sensitivity. Fig. 3 shows the effect of

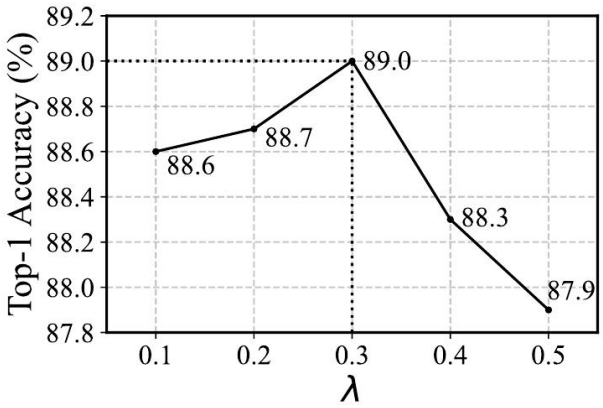

Figure 3. Sensitivity analysis of the contrastive loss weight λ on NTU RGB+D 120 X-Sub accuracy.

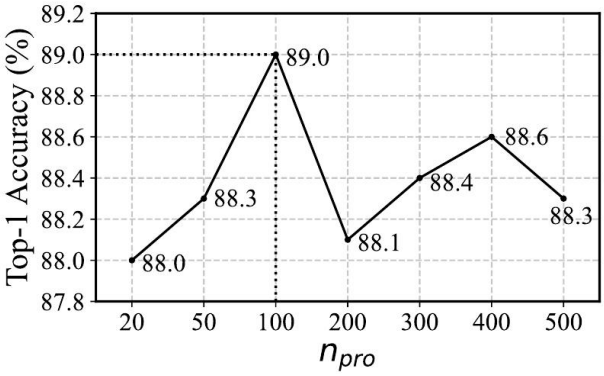

Figure 4. Sensitivity analysis of the prototype memory capacity n_{pro} on NTU RGB+D 120 X-Sub accuracy.

the contrastive loss weight λ. Performance remains stable across a range of values, indicating that ProtoGCN is not overly sensitive to this hyperparameter. Fig. 4 presents the impact of the prototype memory capacity n_{pro}. The model achieves strong performance across various capacity settings, suggesting that a moderate number of prototypes suffices to capture the relevant motion patterns.

Computational efficiency. Table 3 also illustrates that the proposed components introduce only modest overhead relative to their accuracy gains. The MTE module relies on two lightweight 1×1 convolution projections and element-wise operations, adding approximately 2% to the total FLOPs of the backbone. The PRN introduces $2 \times n_{\text{pro}} \times C$ additional learnable parameters—roughly 7.7M at our default settings of $n_{\text{pro}} = 100$ and $C = 384$—but requires only a single matrix multiplication and softmax operation per forward pass, keeping its inference latency negligible. Crucially, because CSCL and its projection head $\mathbf{W}_{cl}$ are training-only components, they contribute zero overhead at test time. Taken

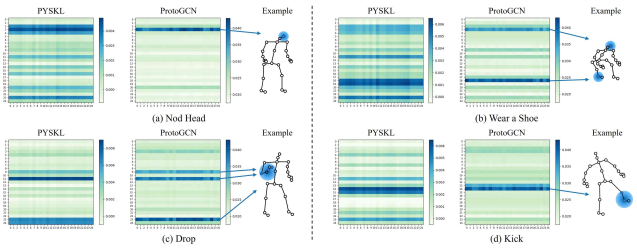

Figure 5. Visualization of learned graph topologies. Compared to PYSKL [7], ProtoGCN produces more structured and discriminative joint relationship patterns through prototype-based reconstruction.

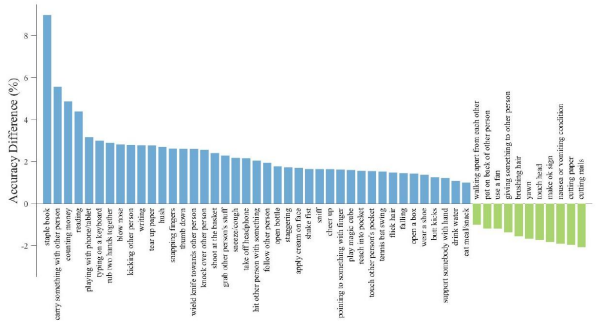

Figure 6. Per-class accuracy difference between ProtoGCN and the baseline on NTU RGB+D 120 X-Sub. Positive values indicate improvement by ProtoGCN.

together, ProtoGCN achieves state-of-the-art accuracy with a parameter overhead that is small relative to the backbone, confirming that prototype-based topology reconstruction is a practical and scalable enhancement for skeleton recognition.

Qualitative analysis. Fig. 5 visualizes the learned graph topologies of PYSKL and ProtoGCN. Our method produces more structured and discriminative joint connectivity patterns, reflecting the capacity of prototype-based reconstruction to capture semantically meaningful relationships. Specifically, the topology patterns learned by ProtoGCN exhibit strong action-dependent locality: high-weight edges cluster around the body segments most informative for each action class. For instance, *hand gesture* actions show concentrated wrist-to-finger connectivity, whereas *lower-body* actions reveal prominent hip-knee-ankle pathways. This prototype-driven selective routing of graph edges constitutes an implicit form of spatial attention, enabling the model to suppress irrelevant joints and amplify the subtle motion signals critical for distinguishing confusable categories. Such structured sparsity is largely absent in standard GCN topologies, where the learned adjacency weights tend to be distributed more uniformly across all joint pairs regardless of action type, contributing to the observed performance gap on the hardest classes. Fig. 6 shows the per-class accuracy difference between ProtoGCN and the baseline. ProtoGCN yields consistent improvements across the majority of classes, with the largest gains concentrated on actions that are inherently difficult to distinguish, further validating the effectiveness of our approach for fine-grained recognition.

Among the action classes benefiting most from ProtoGCN are interaction-heavy activities such as *handshaking*, *hugging*, and *writing*, where the relative positioning of hand joints carries the primary discriminative signal. The prototype memory appears to store recurring joint-interaction patterns as dedicated prototypes, allowing the model to selectively retrieve relevant joint configurations rather than averaging over all possible interpersonal dynamics. In contrast, classes with highly distinctive whole-body kinematics—such as *jumping* or *kicking*—show smaller per-class gains, as these actions are already well-handled by standard graph convolutions. This asymmetry corroborates the central motivation of ProtoGCN: prototype-based topology reconstruction provides the greatest benefit precisely where fine-grained, localized joint relationships constitute the primary discriminative cue.

5. Conclusion

We have presented ProtoGCN, a novel framework for skeleton-based action recognition that introduces a prototypical perspective to address the challenge of distinguishing similar actions. By reconstructing motion topology features through a learnable prototype memory via the Prototype Reconstruction Network (PRN), our method effectively amplifies discriminative joint-level dynamics that conventional GCNs tend to overlook. The Motion Topology Enhancement (MTE) module enriches input representations with intra-sample and inter-sample topology signals, while Class-Specific Contrastive Learning (CSCL) enforces inter-class separability in the feature space. Extensive experiments on NTU RGB+D, NTU RGB+D 120, Kinetics-Skeleton, and FineGYM demonstrate that ProtoGCN achieves state-of-the-art performance across all benchmarks, with particularly pronounced improvements on the most challenging action classes where fine-grained discrimination is critical.

Despite these strengths, ProtoGCN has certain limitations. The prototype memory introduces additional parameters that scale with both the number of prototypes and the feature dimension, and the multi-stream ensemble incurs higher inference cost compared to single-stream baselines. Future work may explore lightweight prototype retrieval mechanisms, online adaptation of class centers beyond momentum updates, and extension of the prototypical perspective to multi-modal settings that combine skeleton data with RGB or depth information.

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